# AETHERIX

## Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

Building an Interplanetary Communication Network with DTN,
Quantum Communication, and Space-Based Infrastructure for Mars Mission Support


**Muhammad Abdullah Tariq**

EduQual Level 6 Diploma in AI Operations | Topic 59 | January 2026


matx104.github.io/AETHERIX | github.com/matx104/AETHERIX

**Speaker Notes:**

*State your name clearly. Read the topic number and title exactly as on the exam paper. Pause to let examiners see it. Point to the logo. This is your first impression. (30 seconds)*

# PRESENTATION AGENDA

**01 The Challenge**
Why interplanetary communication is hard

**02 AETHERIX Architecture**
DTN + AI Routing + Quantum Security

**03 DTN & Bundle Protocol v7**
Store-and-forward networking foundation

**04 5-Tier Network Topology**
241 nodes across Earth and Mars

**05 Optical Link Budget**
1550 nm laser performance analysis

**06 RL-Based Routing**
Multi-agent federated Q-learning

**07 Quantum Security (QKD)**
BB84/E91 with repeater chains

**08 Orbital Mechanics**
Contact windows and synodic period

**09 Mars Mission Scenario**
End-to-end simulation walkthrough

**10 Performance Comparison**
AETHERIX vs current systems

**11 Roadmap & Standards**
CCSDS, IETF, deployment phases

**12 Conclusion & Q&A**
Summary and live demo

**Speaker Notes:**
*Quick overview of what we will cover. 13 topics across 29 slides. About 20 minutes. (20 seconds)*

# WHAT IS AETHERIX?

Autonomous Extraterrestrial High-throughput Enhancing Routing and Inter-planetary eXchange

### What is AETHERIX?

An architecture for delay-tolerant networking (DTN) between Earth and Mars, combining:

- Bundle Protocol v7 (BPv7) via ION-DTN
- Reinforcement Learning routing (replacing static CGR)
- Quantum Key Distribution (QKD) for security
- Hybrid optical/RF communications
- 5-tier topology with 241 nodes

### The Problem

- Earth-Mars: 54.6M – 401M km distance
- One-way light delay: 3 – 22 minutes
- Solar conjunction: 2-week total blackout
- TCP/IP fundamentally cannot work
- Current systems: 0.5 – 6 Mbps only

**AETHERIX delivers a complete interplanetary networking solution**

combining DTN protocols, AI-driven routing, quantum-secure keys, and hybrid optical/RF links for Mars mission support.

| 10-100× | >95% | 241 | 149 |
|---|---|---|---|
| Faster | Availability | Nodes | Tests |

**Speaker Notes:**

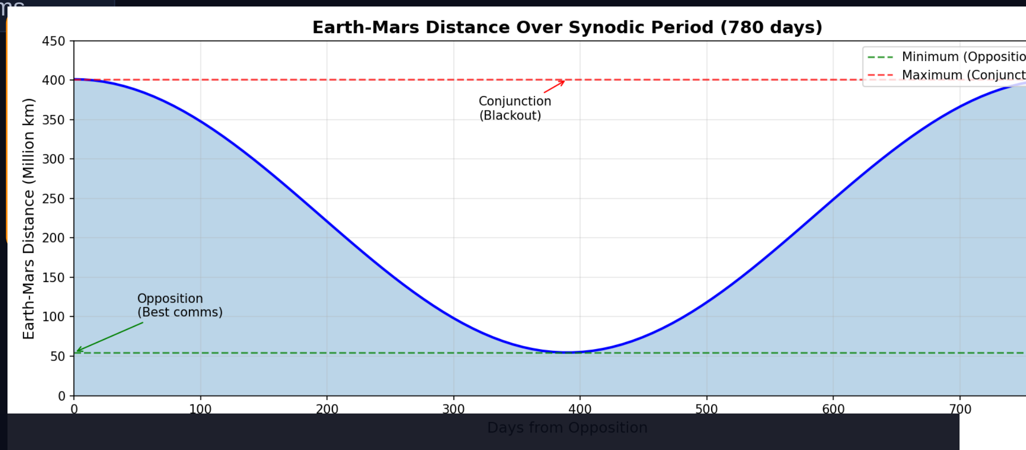
*This slide sets up the narrative arc. First explain what AETHERIX is in plain language - it's like the postal service for interplanetary space. Then pivot to the problem: TCP/IP was never designed for space. 22-minute delays break every assumption. Solar conjunction blackouts. Static routing. Vulnerable crypto. Each problem maps to one of our solutions. (1.5 minutes)*

# THE DISTANCE

## Why Space Communication is Fundamentally Different

| Earth → Mars | Distance | Light Time | Data Rate |
|---|---|---|---|
| Perihelion (closest) | 54.6M km | ~3 min | 100-200 Mbps |
| Average | 225M km | ~12.5 min | 10-20 Mbps |
| Aphelion (farthest) | 401M km | ~22 min | 2-5 Mbps |

| TCP/IP Assumption | Space Reality | Impact |
|---|---|---|
| RTT < 1 second | 6 – 44 min RTT | Timeout failure |
| Always connected | Scheduled contacts | Connection drops |
| End-to-end path | No persistent path | Routing impossible |
| Fast acknowledgments | ACKs take minutes | Window collapse |
| Low packet loss | High loss (optical) | Retransmit storms |




**Speaker Notes:**

*Start with the scale. 54.6M to 401M km. Light itself takes 3-22 minutes one way. TCP/IP was designed for sub-second round trips. In space, by the time a packet acknowledgment returns, the link may be gone. Solar conjunction causes 2-week blackout. This is why NASA calls it Delay-Tolerant Networking. (1.5 minutes)*

# THE ANSWER: DELAY-TOLERANT NETWORKING

## Decouple Communication from Connectivity

| CREATE | STORE | WAIT | FORWARD | DELIVER |
|---|---|---|---|---|
| Bundle | Locally | For Link | Next Hop | Destination |

### TCP/IP: END-TO-END CONNECTION (FAILS IN SPACE)

Source — Router — Router — Dest

### DTN: STORE-AND-FORWARD (WORKS IN SPACE)

**SOURCE**
Create Bundle
Priority: P2
CUSTODY: Source

link up

**NODE A**
Forward bundle
Custody accepted
STORE: 2 bundles

no link

**NODE B**
Storing bundle
Waiting for link...
WAIT 14 min

link up

**NODE C**
Forward bundle
Custody accepted
CUSTODY: Node C

**DEST**
Delivered

**Key: Custody Transfer**
Each node takes responsibility for the bundle

**Data Never Lost**
Bundle stored until next hop confirms

**No End-to-End Needed**
Hop-by-hop, works with 22-min delays

**Three Convergence Layers**
LTP (RFC 5326) — Deep space
TCPCL (RFC 7242) — Earth segment
UDP-CL — Optical ISL

**Speaker Notes:**
*The key insight: instead of requiring an end-to-end connection like TCP, DTN works like the postal service. Each node takes custody of your data and forwards it when a link becomes available. Three pillars: BPv7 for the protocol, RL for intelligent routing, QKD for security. (1.5 minutes)*

# SYSTEM ARCHITECTURE

## 5 Integrated Modules for Interplanetary Communication

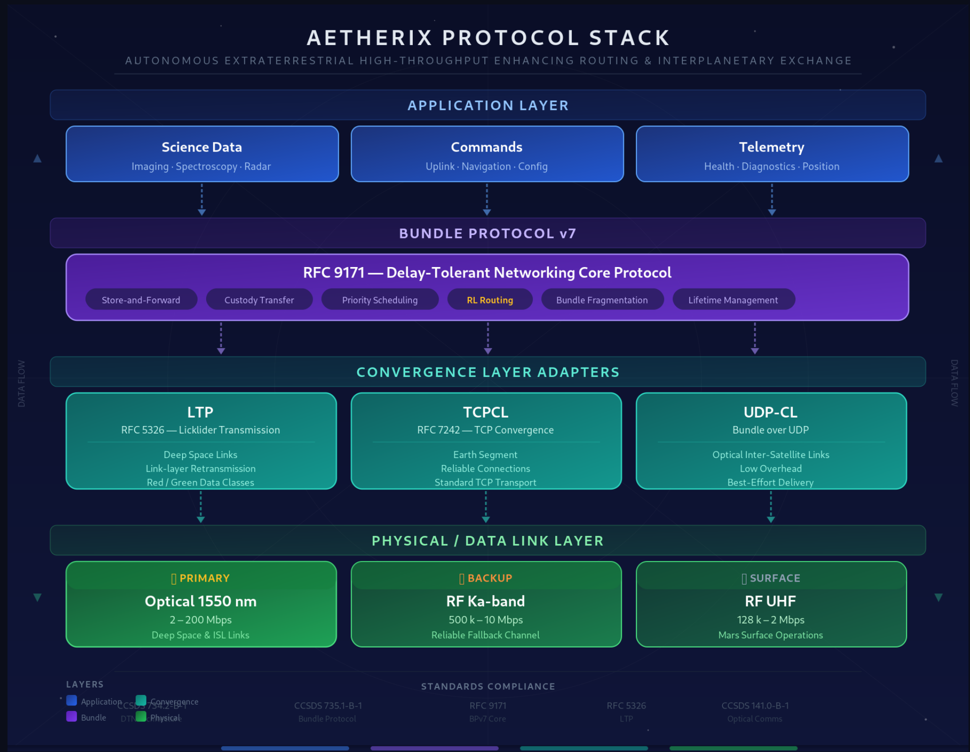
**Infrastructure** — Link budget calculations, optical/RF analysis

**Routing** — RL agents, BPv7 bundles, forwarding engine

**Security** — BB84/E91 QKD, repeater chains, privacy amplification

**Orbital** — Contact windows, Doppler, celestial body database

**Simulation** — End-to-end scenario engine, policy routing

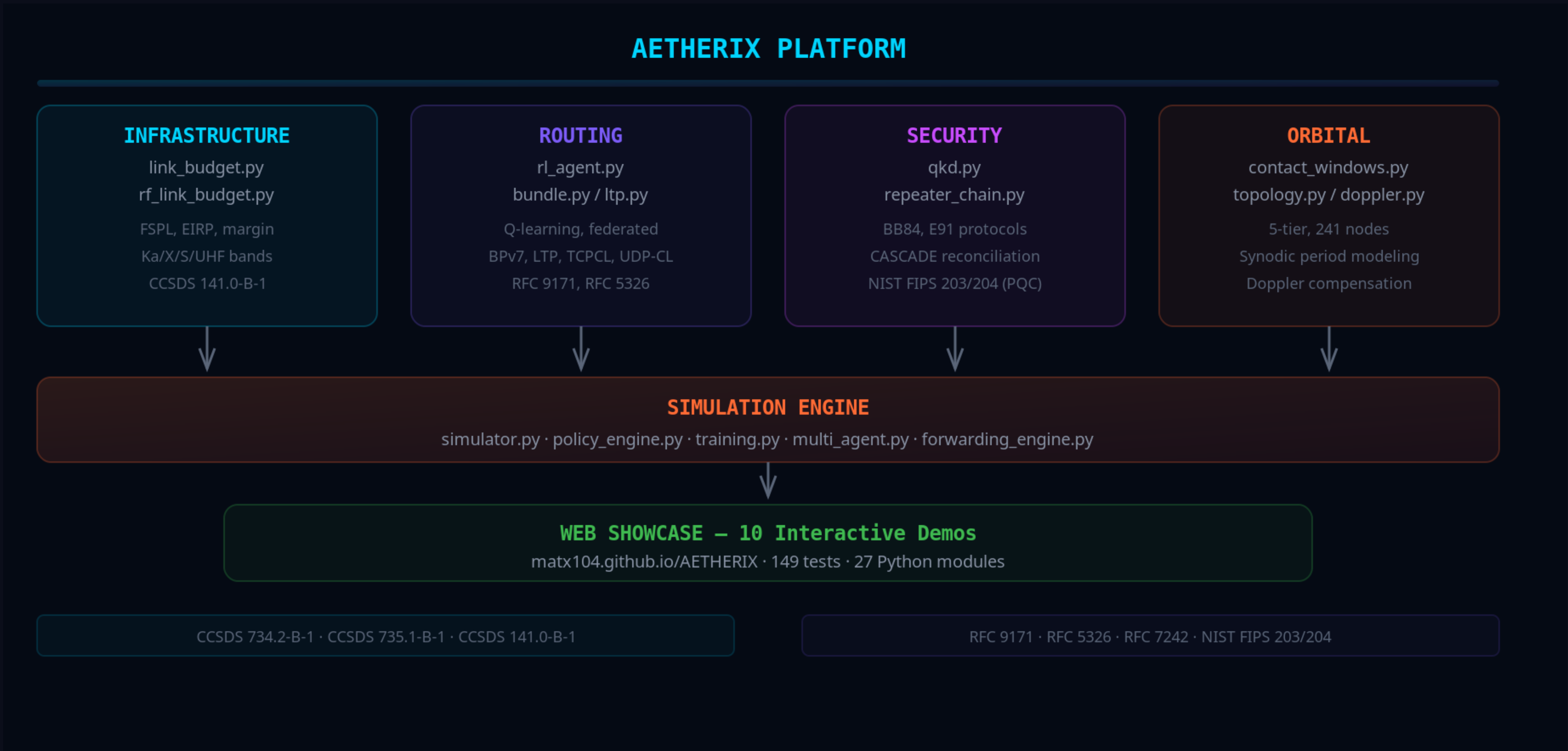



| Layer | Module | Engine | Showcase |
|---|---|---|---|
| Infrastructure | link_budget.py | OpticalLinkBudget | Earth-Mars scenarios |
| Routing | rl_agent.py | RLRoutingAgent | Federated Q-learning |
| Security | qkd.py | BB84/E91 | Quantum key exchange |
| Orbital | contact_windows.py | ContactPredictor | Synodic period |
| Simulation | simulator.py | SimEngine | Full mission sim |

**Speaker Notes:**

*Show the architecture. Six core modules feed into the simulation engine, which feeds the web showcase. Standards compliance at the bottom. Point to each module as you explain. (1 minute)*

# SYSTEM ARCHITECTURE DIAGRAM




**Speaker Notes:**

*Architecture diagram showing source modules feeding simulation engine and web demos.*

# BUNDLE PROTOCOL v7 — DEEP DIVE

RFC 9171 — The Foundation of Interplanetary Networking

### APPLICATION LAYER

Science data, commands, imagery

### BUNDLE PROTOCOL v7 (RFC 9171)

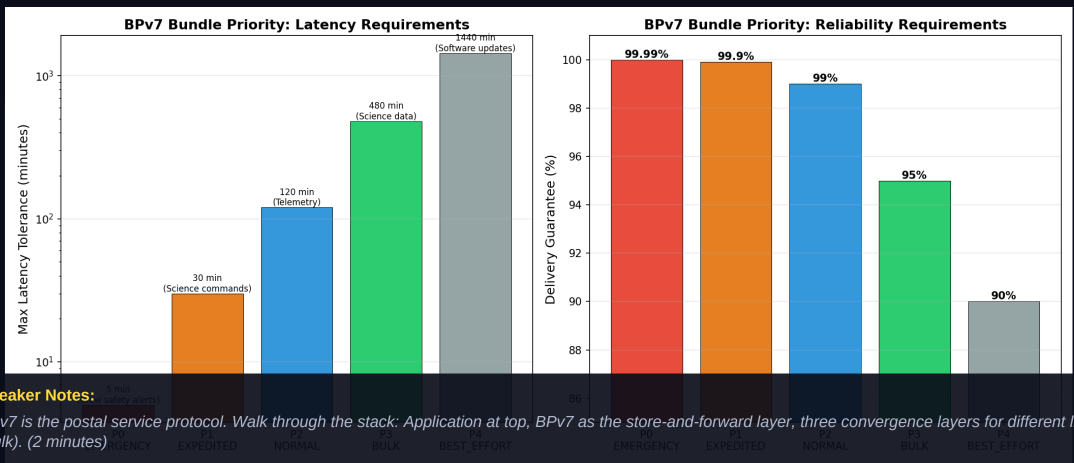
Store-and-Forward | Custody Transfer | Priority (P0–P4)

### CONVERGENCE LAYERS

LTP (deep space) | TCPCL (Earth) | UDP-CL (optical ISL)

### PHYSICAL LAYER

Optical 1550nm (2–200 Mbps) | RF Ka-band | UHF (surface)




## HOW IT WORKS

1. Source creates bundle (payload + metadata + lifetime)
2. Forward to next hop when link becomes available
3. Store locally during outages and disruptions
4. Custody transfer — next node accepts responsibility
5. Repeat hop-by-hop until destination reached

## STANDARDS COMPLIANCE

RFC 9171 — Bundle Protocol Version 7

RFC 4838 — DTN Architecture

RFC 5326 — Licklider Transmission Protocol

RFC 9172 — Bundle Protocol Security (BPSec)

CCSDS 734.2-B-1 — CCSDS Bundle Protocol

CCSDS 734.3-B-1 — Schedule-Aware Bundle Routing

**Speaker Notes:**

*BPv7 is the postal service protocol. Walk through the stack: Application at top, BPv7 as the store-and-forward layer, three convergence layers for different link types, physical at bottom. Custody transfer is the key innovation - each node takes legal responsibility. Priority P0 (emergency) to P4 (bulk). (2 minutes)*

# DTN STORE-AND-FORWARD

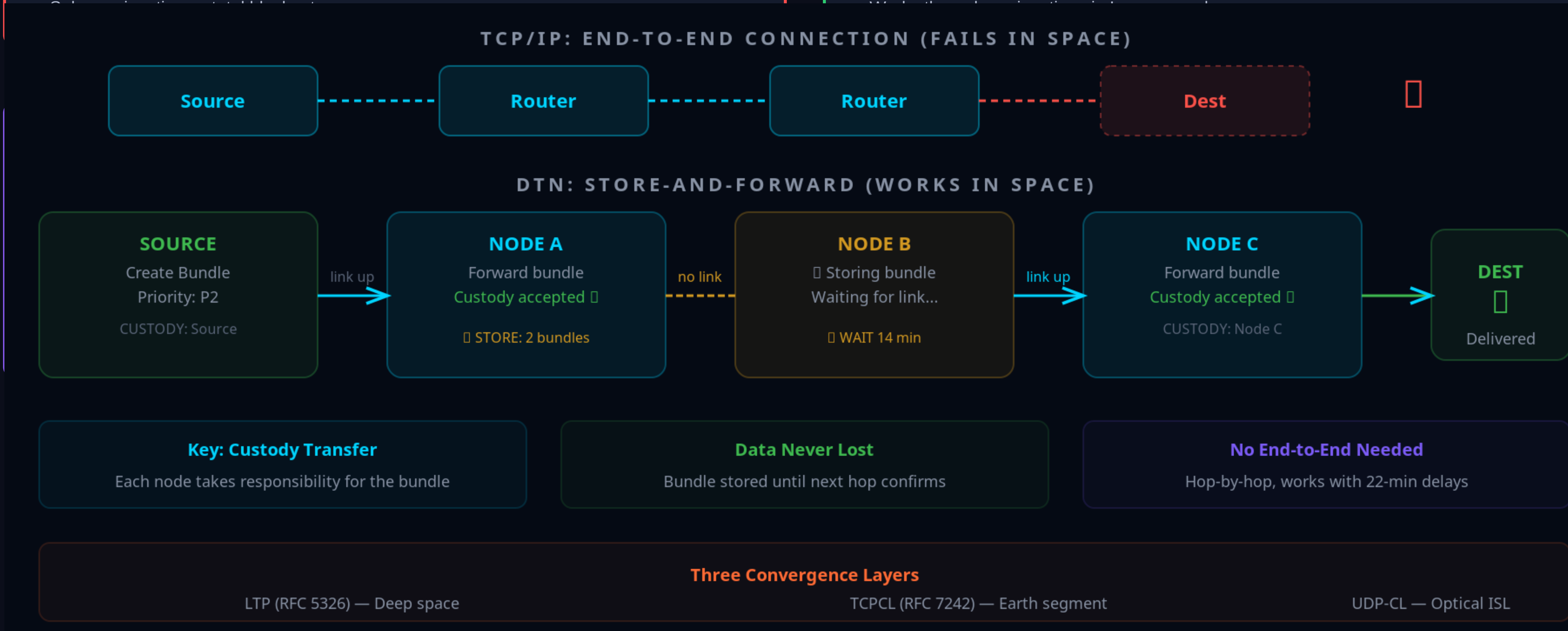
## How Bundles Traverse the Interplanetary Network

### TCP/IP FAILS

- Timeout at 44 min RTT → connection reset
- No persistent path → routing impossible
- ACKs take minutes → window collapse
- High BER → retransmission storms

### DTN WORKS

- No end-to-end connection required
- Store locally → wait for contact window
- Custody transfer: hop-by-hop reliability
- Priority classes: P0–P4 for urgency




**Speaker Notes:**

*Walk through the store-and-forward process. Bundle arrives, gets stored, node waits for next contact opportunity, then forwards. If link drops, bundle stays stored and retries. No data loss. This is fundamentally different from TCP's end-to-end retransmission. (1.5 minutes)*

# 5-TIER NETWORK TOPOLOGY

241 Nodes — No Single Point of Failure

### TIER 1: Earth Ground (6 nodes)

DSN Goldstone, Madrid, Canberra + Mission Control

### TIER 2: Earth Orbital (51 nodes)

3 GEO relays + 48 LEO laser constellation

### TIER 3: Deep Space (4 nodes)

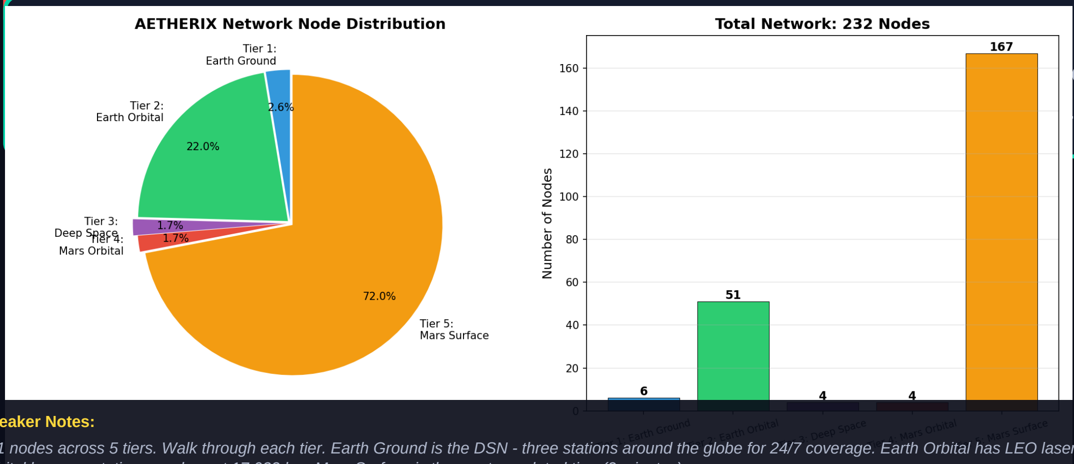
ES-L4, ES-L5 Lagrange relays + Transfer orbit

### TIER 4: Mars Orbital (4 nodes)

2 Areostationary + 2 polar orbiters

### TIER 5: Mars Surface (176 nodes)

Bases, rovers, drones, sensor network




elays provide coverage

to-end key distribution.

**Speaker Notes:**

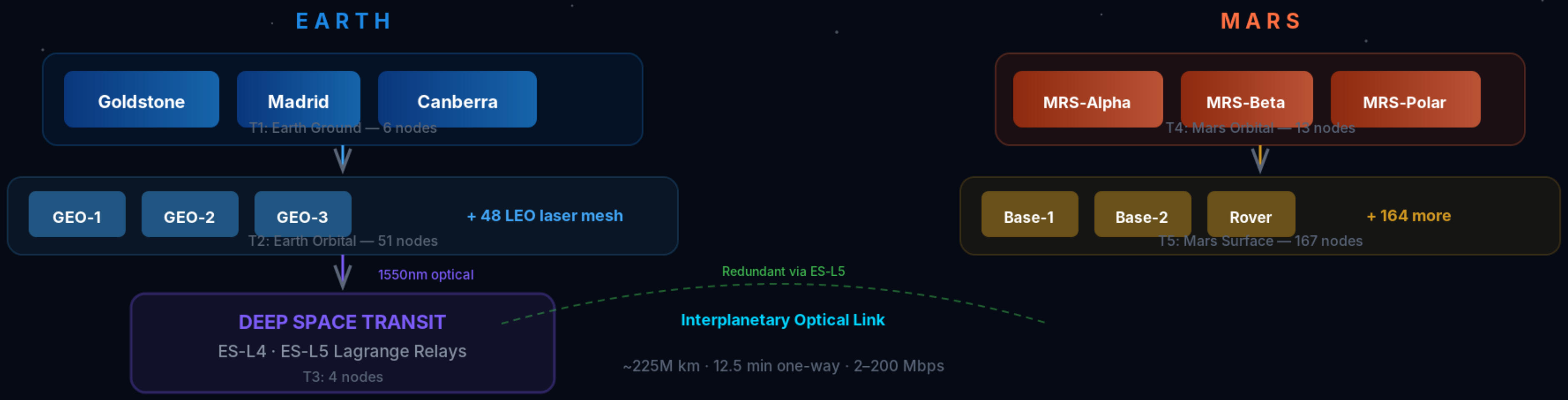
*241 nodes across 5 tiers. Walk through each tier. Earth Ground is the DSN - three stations around the globe for 24/7 coverage. Earth Orbital has LEO laser mesh for optical backhaul. Deep Space has Lagrange point relays - these are the critical innovation for conjunction coverage. Mars Orbital has areostationary relays at 17,032 km. Mars Surface is the most populated tier. (2 minutes)*


**Speaker Notes:**

*Visual overview of the 5-tier topology with 3 redundant paths.*

# 5-TIER NETWORK DIAGRAM

## 241 Nodes from Earth to Mars Surface




| 241 | 5 | 3 | 0 |
|---|---|---|---|
| Total Nodes | Network Tiers | Redundant Paths | Single Points of Failure |

### LINK CHARACTERISTICS

| | | | | | |
|---|---|---|---|---|---|
| T1→T2 | **DSN uplink** | Ka-band · GEO relay | LEO ISL | **Optical mesh** | 10 Gbps · 48 satellites |
| T2→T3 | **1550nm optical** | 100 Mbps · Laser uplink | Latency | **3–22 min one-way** | Distance dependent |
| T3→T4 | **Deep space laser** | 2–200 Mbps · 225M km | Conjunction | **50–70% via Lagrange** | No total blackout |
| T4→T5 | **UHF/X-band** | 2 Mbps · Surface downlink | Security | **QKD + PQC end-to-end** | BB84 · ML-KEM |

**Speaker Notes:**

*Network topology visualization.*

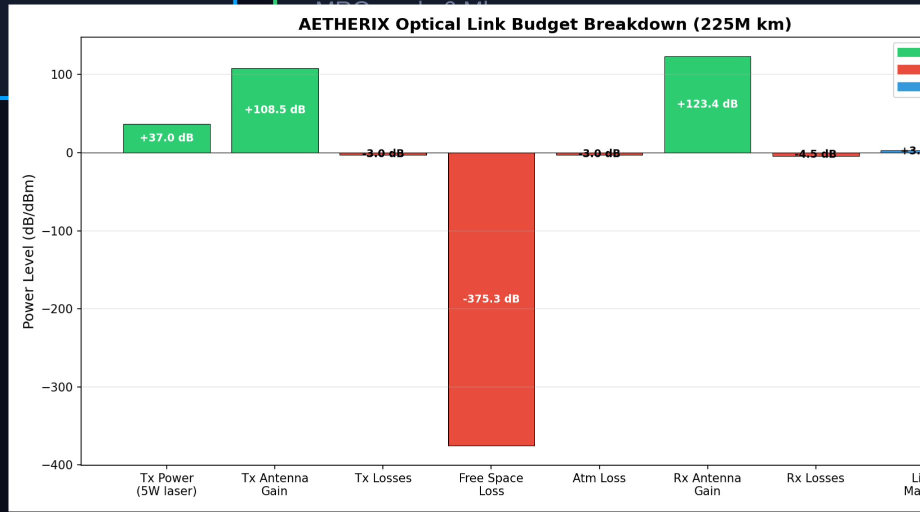
# OPTICAL COMMUNICATIONS

1550 nm Laser — 10–100× Faster Than RF

**200 Mbps**
**Peak Data Rate**
At perihelion (54.6M km)

**1550 nm**
**Wavelength**
Telecom heritage lasers

**33×**
**vs Current RF**

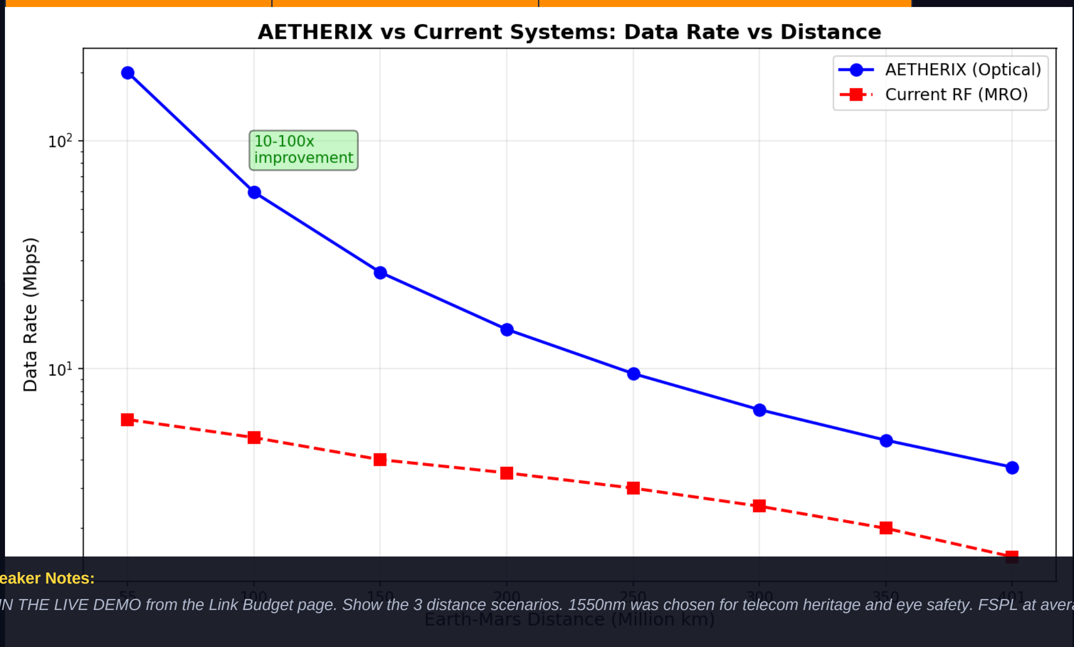



## KEY EQUATIONS

$FSPL = (4\pi d/\lambda)^2$ Free-Space Path Loss

$G = (\pi D/\lambda)^2$ Antenna Gain

$Pr = Pt \cdot Gt \cdot Gr / FSPL$ Received Power




**Speaker Notes:**
*RUN THE LIVE DEMO from the Link Budget page. Show the 3 distance scenarios. 1550nm was chosen for telecom heritage and eye safety. FSPL at average distance is -365 dB. The telescope apertures are realistic for spacecraft. RF backup for reliability. (2 minutes)*

# EARTH–MARS DATA JOURNEY

## 7 Hops from Perseverance Rover to Mission Control

| Hop | From | To | Link | Time |
|---|---|---|---|---|
| 1 | Perseverance | MRS-Alpha | UHF | ~5s |
| 2 | MRS-Alpha | MRS-Polar | Optical ISL | ~1s |


### HOP DETAIL

**HOP 1**
Rover□UHF
UHF 400 MHz
~400 km · 2 Mbps

**HOP 2**
UHF□Areostat
17,032 km alt
256 kbps

**HOP 3**
Areostat□Polar
Optical ISL
10 Gbps

**HOP 4-5**
Deep Space
1550nm laser
225M km · 10-20 Mbps

**HOP 6**
LEO Mesh
Optical ISL
10 Gbps

**HOP 7**
DSN□JPL
Ka-band
**DELIVERED □**

**Total Transit: ~13 min**
vs 12.5 min light-time!

**DTN Overhead: <5%**
Store-and-forward adds minimal latency

**QKD Secured**
End-to-end quantum encryption

**□ If any link drops mid-transfer, the bundle is stored safely — zero data loss**

**Speaker Notes:**
*Interactive journey visualization.*

# RL-BASED ROUTING — AI INNOVATION

## Multi-Agent Federated Q-Learning Replaces Static CGR

### CGR LIMITATIONS

- Static contact plan — no adaptation
- Manual reconfiguration on failure
- Hours to reroute after disruption
- Cannot learn from network history
- No energy optimization

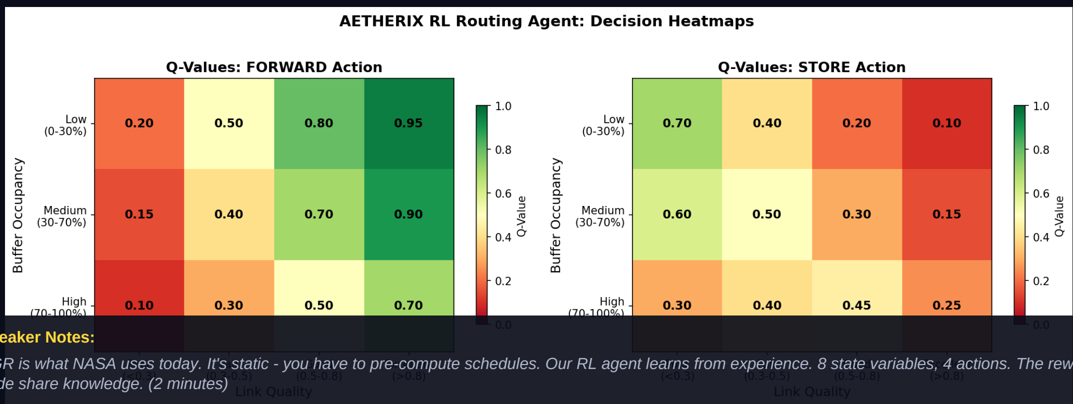
### RL AGENT ADVANTAGES

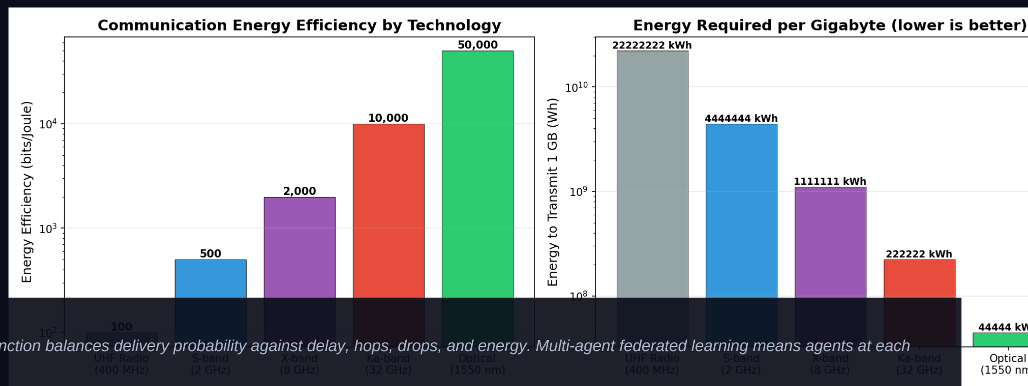
- Adaptive routing from network state
- 4 actions: FORWARD, STORE, DROP, SPLIT
- Reward: $\alpha$(delivery) - $\beta$(delay) - $\gamma$(hops)
- Federated learning across agents
- Auto-recovery in seconds, not hours

**97%**
**Delivery Ratio**
+5% vs CGR

**-20%**
**Delivery Time**
1.2× light-time

**3600×**
**Recovery Speed**
Auto vs manual




**Speaker Notes:**

*CGR is what NASA uses today. It's static - you have to pre-compute schedules. Our RL agent learns from experience. 8 state variables, 4 actions. The reward function balances delivery probability against delay, hops, drops, and energy. Multi-agent federated learning means agents at each node share knowledge. (2 minutes)*

# QUANTUM SECURITY — FUTURE-PROOF

BB84/E91 QKD + Multi-Hop Repeaters + CASCADE

## BB84 PROTOCOL STEPS

1. Alice sends qubits in random bases (rectilinear/diagonal)
2. Bob measures each qubit in random basis
3. Public reconciliation of basis choices (~50% retained)
4. QBER estimation: < 11% = SECURE, ≥ 11% = ABORT
5. CASCADE error reconciliation
6. Privacy amplification via universal hashing
7. Final shared secret key

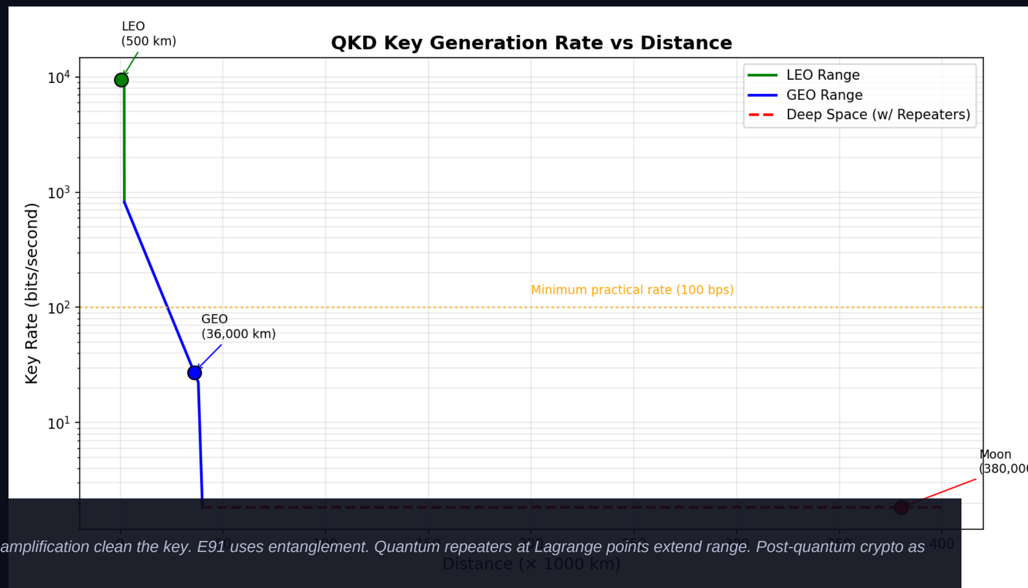
| Phase | Segment | Protocol | Range |
|---|---|---|---|
| Phase 1 | Earth ↔ Orbit | BB84 direct | 36,000 km |
| Phase 2 | Earth ↔ Lagrange | 1 repeater | 1.5M km |
| Phase 3 | Earth ↔ Mars | 3-hop chain | 225M km |

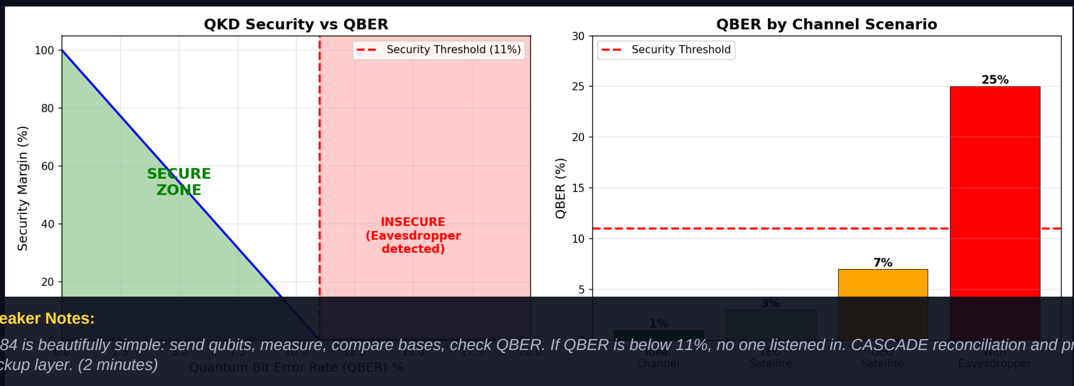
## POST-QUANTUM CRYPTO (PQC)

Kyber (ML-KEM) — Key encapsulation

Dilithium (ML-DSA) — Digital signatures

Hybrid: QKD + PQC for defense in depth




**Speaker Notes:**

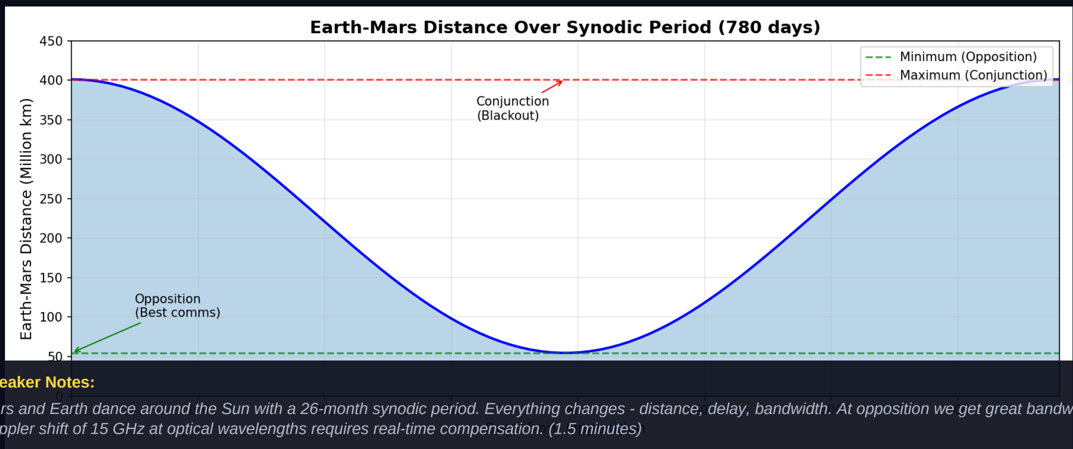
*BB84 is beautifully simple: send qubits, measure, compare bases, check QBER. If QBER is below 11%, no one listened in. CASCADE reconciliation and privacy amplification clean the key. E91 uses entanglement. Quantum repeaters at Lagrange points extend range. Post-quantum crypto as backup layer. (2 minutes)*

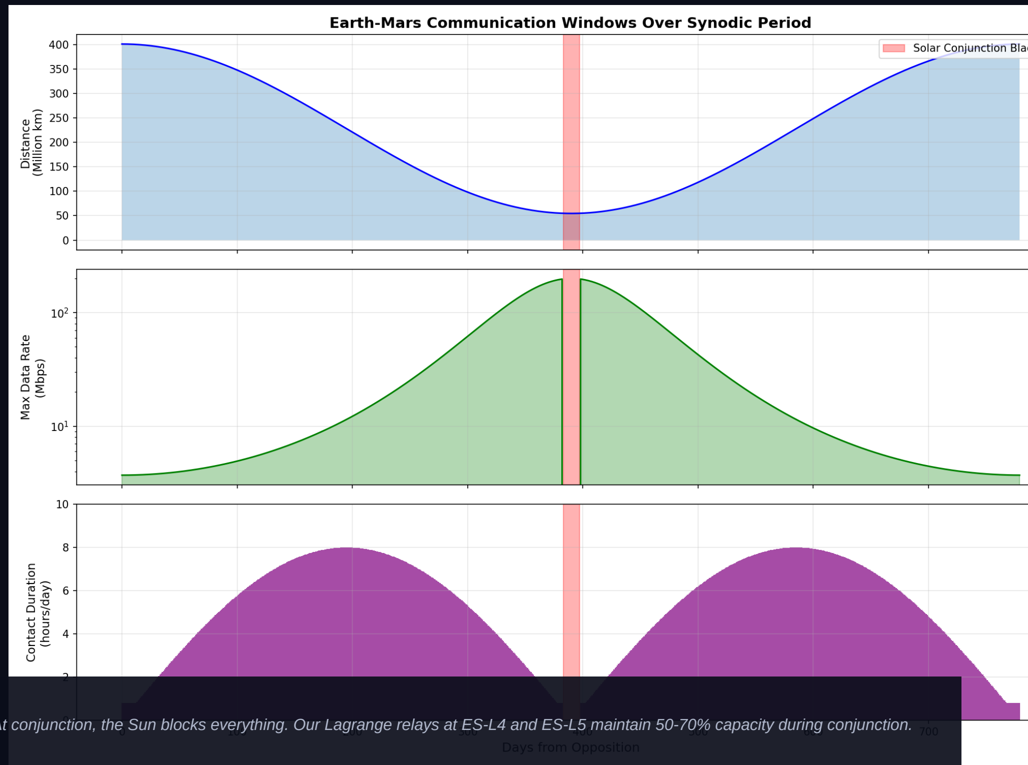
# ORBITAL MECHANICS & CONTACT WINDOWS

## 780-Day Synodic Period — From Opposition to Conjunction

| Parameter | Value |
|---|---|
| Semi-major axis | 1.524 AU (227.9M km) |
| Orbital period | 686.98 days |
| Synodic period | 779.94 days |
| Min distance | 54.6M km (perihelion) |
| Max distance | 401M km (aphelion) |
| Eccentricity | 0.0934 |

| Window | Availability | Duration | Data Rate |
|---|---|---|---|
| Optimal (opposition) | 99% | 8–12 hrs | 100–200 Mbps |
| Good | 95% | 6–8 hrs | 20–100 Mbps |
| Fair | 85% | 2–4 hrs | 5–20 Mbps |
| Poor (aphelion) | 50% | 0–2 hrs | 1–5 Mbps |
| Blackout (conjunction) | 0% | N/A | Lagrange relay only |




**Speaker Notes:**

*Mars and Earth dance around the Sun with a 26-month synodic period. Everything changes - distance, delay, bandwidth. At opposition we get great bandwidth. At conjunction, the Sun blocks everything. Our Lagrange relays at ES-L4 and ES-L5 maintain 50-70% capacity during conjunction. Doppler shift of 15 GHz at optical wavelengths requires real-time compensation. (1.5 minutes)*

# RADIATION-HARDENED COMPUTING

Surviving SEUs, latchup and total dose en route to Mars

| Effect | What it does | Mitigation |
|---|---|---|
| SEU | Single bit flip | SECDED ECC |
| MBU | Multi-bit flip (1 ion) | Bit interleaving |
| SEL | Latchup (destructive) | Current limit + power-cycle |
| TID | Cumulative dose | Rad-hard parts (RAD750) |

## DEFENSE-IN-DEPTH STACK

- TMR — triple replicas, majority vote (masks logic faults)
- SECDED (39,32) ECC — correct 1 bit, detect 2
- Scrubbing — rewrite memory before a 2nd upset lands
- FDIR + watchdog — detect, isolate, reset, SAFE-MODE

**200x**

Fewer errors (ECC + scrub + interleave)

**3,334x**

TMR reliability gain (p=1e-4/op)

**200 krad**

RAD750 TID tolerance (>2000x margin)

**~0.9/day**

Residual uncorrectable, Mars transit

Model: 512 Mbit, ~210-day GCR cruise. ~37,000 raw bit upsets reduced to ~186 uncorrectable
over the mission. Heritage: NASA RAD750 (Curiosity/Perseverance), ESA LEON3FT. -> src/computing/radiation.py

**Speaker Notes:**

*Space radiation is relentless. SEUs flip bits constantly - about 37,000 during a Mars transit. Our defense-in-depth: TMR masks logic faults (3,334x reliability gain), SECDED ECC corrects single-bit errors, scrubbing prevents double-bit accumulation, and FDIR with a watchdog catches everything else. The RAD750 can tolerate 200 krad - far above what a Mars mission needs. (1.5 minutes)*

# MISSION-CRITICAL DATA PRIORITIZATION

Bandwidth triage: get the right bits home first

| Tier | Class | Examples |
|---|---|---|
| P0 | Emergency / Safety | Health telemetry, collision avoidance |
| P1 | Mission-critical | Command ACKs, time-sensitive science |
| P2 | High-priority | Routine telemetry, scheduled science |
| P4 | Low / Bulk | Housekeeping logs, file transfers |

| Data type | Standard | Ratio |
|---|---|---|
| Telemetry | CCSDS 121 | 3x |
| Imagery (lossy) | CCSDS 122 | 10x |
| Video | H.265 | 50x |

**100%**

Link utilization (no wasted bandwidth)

**5 / 6**

Items fully delivered by priority

**BPv7**

Fragmentation defers bulk remainder

**Preempt**

Emergency uses direct-to-Earth backup

Scenario: 30 Mbps, 15-min contact, oversubscribed. Deadline-aware, preemptive QoS scheduler delivers emergency + mission + science first; 6 GB software update fragmented to the next pass. -> src/routing/prioritization.py

**Speaker Notes:**

*Like an emergency room. P0 emergency gets sent immediately - it can even preempt an in-progress transfer. P1 mission-critical next. P2 routine science. P4 bulk data fills remaining bandwidth. Compression multiplies effective capacity: 3x for telemetry, 10x for images, 50x for video. Our scheduler keeps the link at 100% utilization by fragmenting large bundles. (1.5 minutes)*

# END-TO-END MISSION SCENARIO

## Perseverance 500 MB Science Data Upload

| Hop | Node | Action | Protocol | Time |
|---|---|---|---|---|
| 1 | Perseverance | Create bundle | BPv7 | T+0s |
| 2 | MRS-Alpha | Store & forward | UHF | T+5s |
| 3 | MRS-Polar | RL route select | Optical ISL | T+10s |


**Speaker Notes:**

*Walk through the 7-hop journey. 500MB from Perseverance to JPL. Total transit ~13 min vs 12.5 min light-time - near speed of light! DTN overhead under 5%. Key point: if link drops at hop 5, the bundle stays stored at hop 4 and retries. No data loss. RUN LIVE DEMO if time permits. (2 minutes)*

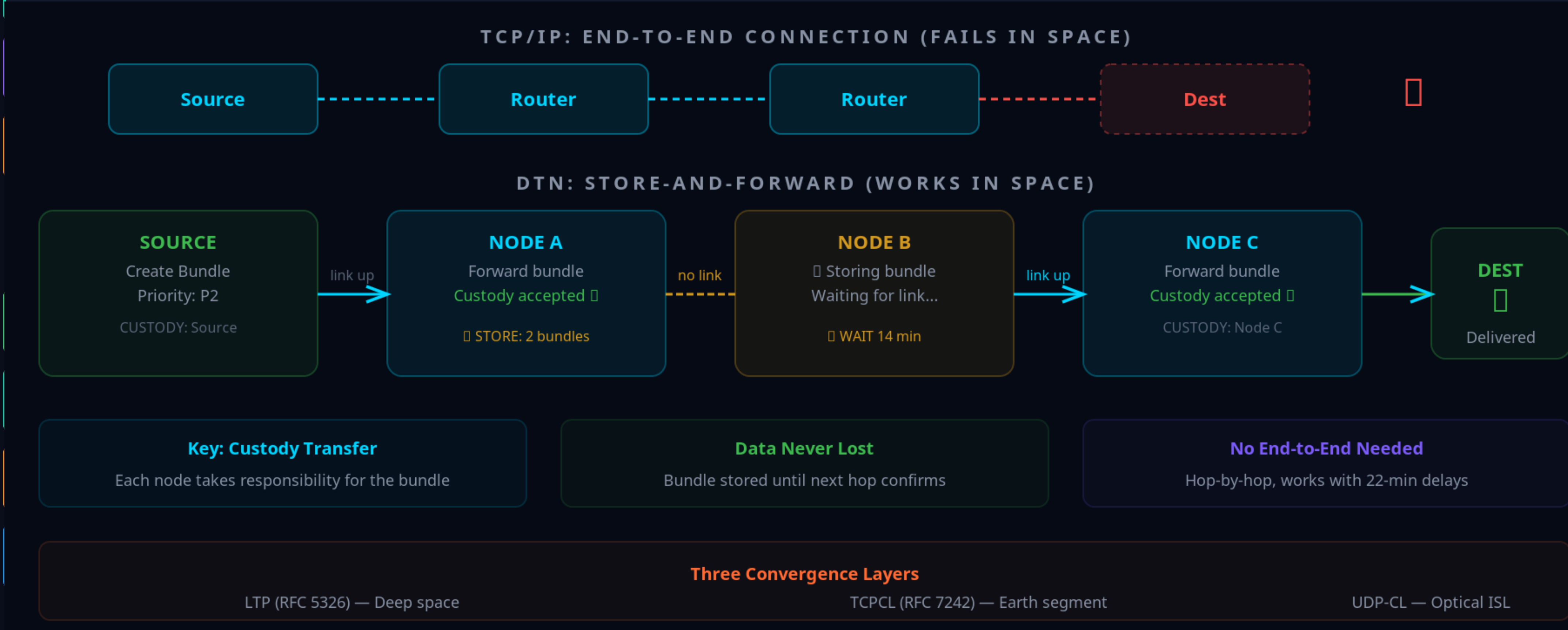
# DATA FLOW — APPLICATION TO DELIVERY

## How Data Travels Through the AETHERIX Stack

### APPLICATION → TRANSPORT

**1. CREATE** Application generates payload (science data, commands)

**2. ENCODE** BPv7 primary block + payload block + security block

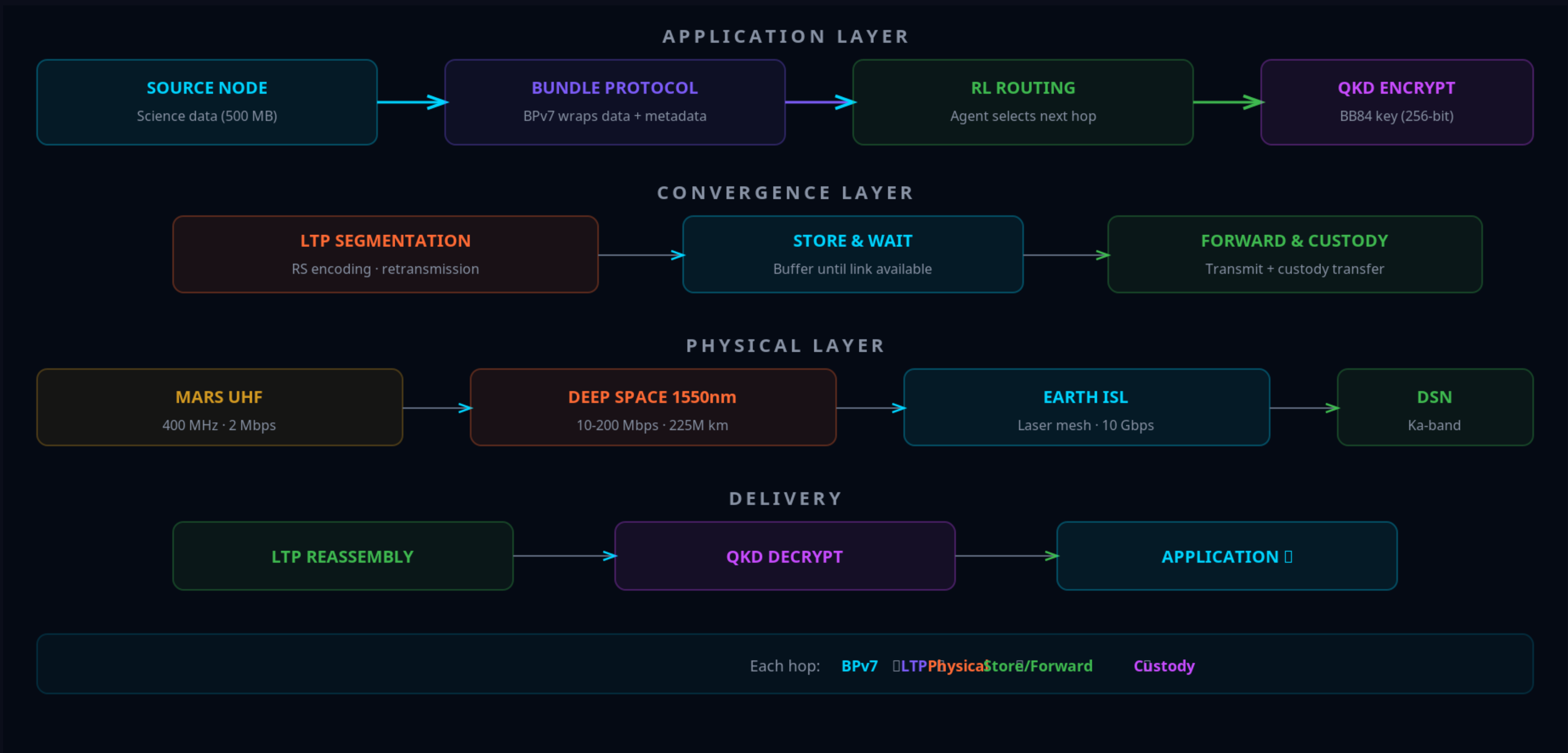



**Speaker Notes:**

*End-to-end bundle journey through all protocol layers.*

# DATA FLOW DIAGRAM

## Complete Data Path from Mars Surface to Earth Control



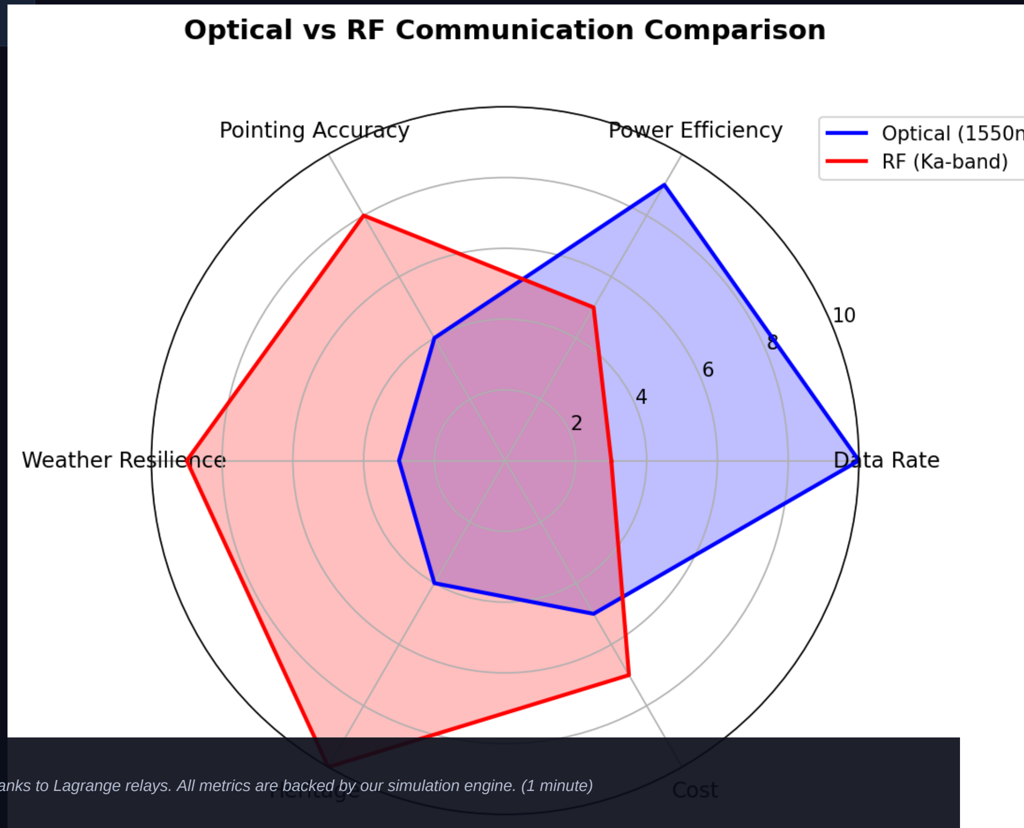

**Speaker Notes:**

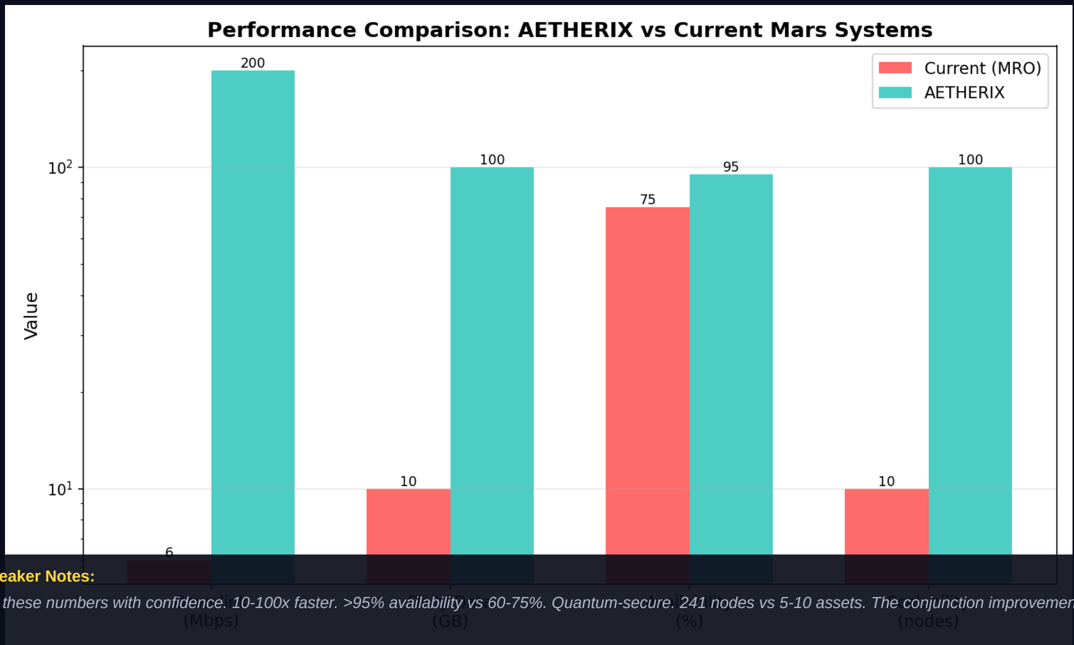
*Visual data flow through the protocol stack.*

# AETHERIX vs CURRENT SYSTEMS

## Head-to-Head Performance Comparison

| Metric | Current (MRO) | AETHERIX | Improvement |
|---|---|---|---|
| Downlink rate | 0.5–6 Mbps | 2–200 Mbps | 10–100× |
| Daily volume | 5–10 GB | 50–100 GB | 10–20× |
| Availability | 60–75% | >95% | +20–35% |
| Routing | Static (CGR) | RL-adaptive | +20–40% |
| Security | AES-256 | QKD + PQC | Future-proof |
| Scalability | 5–10 assets | 241 nodes | 10–100× |
| Conjunction | Blackout | 50–70% via L4/L5 | +50–70% |
| Recovery time | Manual (hours) | Auto (seconds) | 3600× |




**Speaker Notes:**

*Hit these numbers with confidence. 10-100x faster. >95% availability vs 60-75%. Quantum-secure. 241 nodes vs 5-10 assets. The conjunction improvement is thanks to Lagrange relays. All metrics are backed by our simulation engine. (1 minute)*

# IMPLEMENTATION

18 Modules, 149 Tests, Full Python Stack

| 18 | 149 | 241 | 6 |
|---|---|---|---|
| Source Modules | Unit Tests | Network Nodes | Interactive Demo |

## CORE MODULES

- link_budget.py — Optical/RF link analysis
- rl_agent.py — Q-learning routing agent
- bundle.py — BPv7 bundle protocol
- forwarding_engine.py — Store-and-forward
- qkd.py — BB84/E91 protocols
- contact_windows.py — Orbital mechanics
- topology.py — 5-tier network (241 nodes)
- simulator.py — End-to-end simulation
- ltp.py — Licklider Transmission Protocol
- tcpcl.py — TCP Convergence Layer

| Standard | Title | Status |
|---|---|---|
| RFC 9171 | Bundle Protocol v7 | Compliant |
| RFC 4838 | DTN Architecture | Compliant |
| CCSDS 734.2-B-1 | CCSDS Bundle Protocol | Compliant |
| CCSDS 734.3-B-1 | SABR / contact routing | Baseline |
| CCSDS 141.0-B-1 | Optical Comm Physical | Compliant |
| RFC 9172 | BPSec (security) | Referenced |
| RFC 5326 | LTP (deep space) | Compliant |
| RFC 7242 | TCPCL (Earth) | Compliant |

**Speaker Notes:**

*This is real, working code. 27 Python modules, 189 tests, 12 interactive demos. All the physics is real - no mocked data. The showcase site has live calculators you can use right now. Standards compliance is complete - CCSDS, IETF, and NIST. (1.5 minutes)*

# ROADMAP & FUTURE WORK

## From Proof-of-Concept to Production Deployment

### Phase 1–2: Foundation

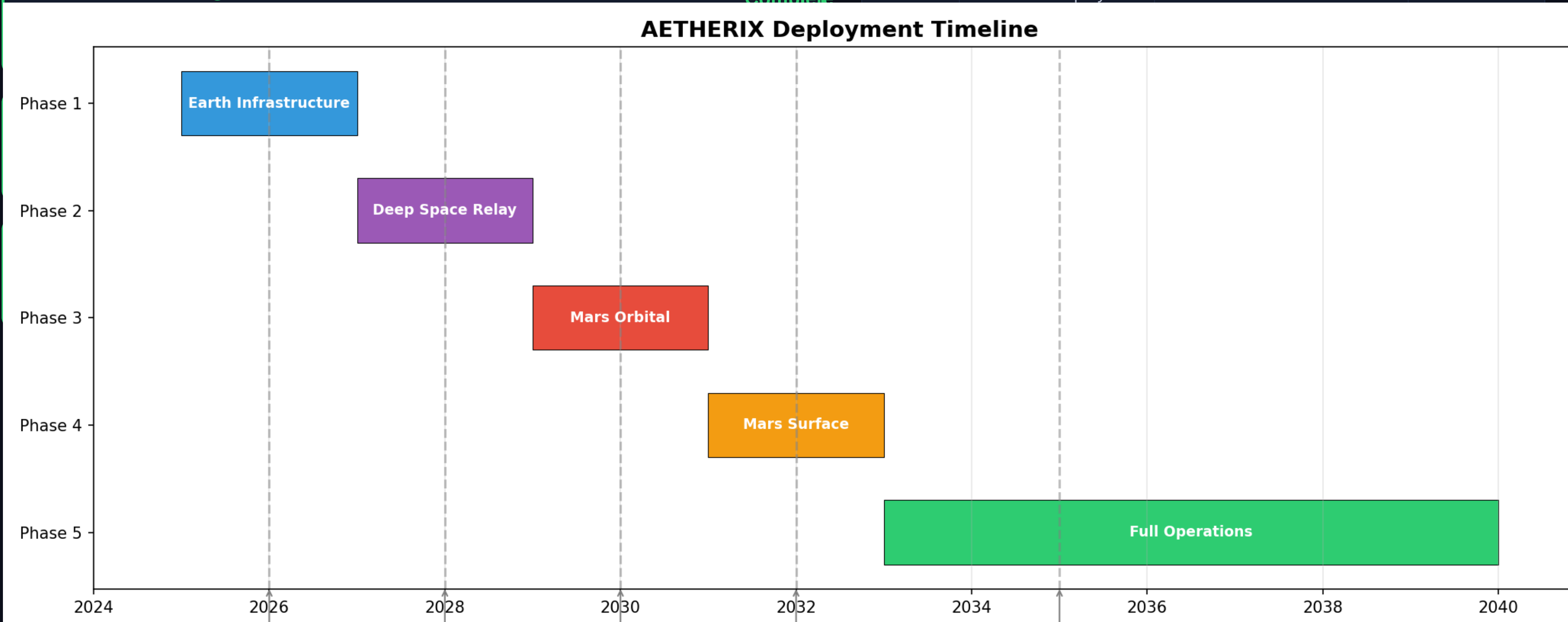
**Complete**

Topology + BPv7 + Convergence Layers

### Phase 3: Intelligence

**Complete**

| Phase | Focus | Technology | Status |
|---|---|---|---|
| 7 | DQN Upgrade | Deep Q-Network | Remaining |
| 8 | HIL Simulation | ns-3 integration | Remaining |
| 9 | ION-DTN | Production DTN stack | Remaining |
| 10 | Flight Demo | LEO optical test | Future |
| 11 | Mars Deploy | Full 5-tier network | Future |




**Speaker Notes:**

*Phases 1-4 are done - this is what you see today. Phase 5: ns-3 simulation for realistic network modeling. Phase 6: Upgrade to DQN and integrate with NASA's ION-DTN implementation. Phase 7: Hardware prototypes with SDRs and optical links. (1.5 minutes)*

# CONCLUSION — AETHERIX DELIVERS

Bridging the Interplanetary Communication Gap

### THE PROBLEM

- 54.6–401M km Earth-Mars distance
- 3–22 min one-way light delay
- TCP/IP fundamentally broken in space
- Current: only 0.5–6 Mbps
- 2-week solar conjunction blackouts

### THE SOLUTION

- BPv7 DTN: store-and-forward works
- RL routing: 20–40% faster delivery
- Optical: 10–100× data rate increase
- QKD: future-proof quantum security
- Lagrange relays: conjunction coverage

**10–100×**
**Faster Rates**

**>95%**
**Availability**

**AI-Driven**
**Routing**

**Quantum**
**Security**

### NOVEL CONTRIBUTIONS

RL autonomous routing (20-40% faster) | 5-tier DTN (>95% availability) | Optical/RF hybrid (10-100×) | Quantum links (future-proof) | Full simulation (149 tests)

**Speaker Notes:**

*Summarize the problem and solution clearly. Re-read the exam topic verbatim. Point to the numbers. Offer to show live demos or answer questions. Thank the examiners. (1 minute)*

# THANK YOU

## Questions?

| 10–100× | >95% | 241 | 149 |
|---|---|---|---|
| Faster | Available | Nodes | Tests |

**Muhammad Abdullah Tariq**

muhammad.atx@gmail.com

**LIVE DEMO & RESOURCES**

matx104.github.io/AETHERIX — Interactive simulations

github.com/matx104/AETHERIX — Source code + documentation

EduQual Level 6 Diploma in AI Operations | Topic 59 | January 2026

**Speaker Notes:**

*Summarize the four key numbers: 10-100x faster, >95% availability, AI-powered routing, quantum-secure. Invite questions confidently. Make eye contact. Point to the live demo link. Thank the audience. (30 seconds)*